

conversions

decimal

binary

octal

hexadecimal

1.decimal to bin ,oct,hex

2.binary

3.octal

4.hexadecimal

Decimal System: The decimal system is the most extensively used number system. This is a base ten numbering system.
A number is represented by 10 numbers (0-9) in this system.

Binary Number System: The binary number system is a base two number system.
Because computers can only understand binary numbers, the binary system is utilized (0 and 1).
Binary numbers have the prefix 0b

Octal system: Octal system is a base 8 numbering system.
octal numbers have the prefix 0o

Hexadecimal Number System: The Hexadecimal number system is a base 16 number system
hexadecimal numbers have the prefix 0x.

Convert **Decimal** to Binary, Octal and Hexadecimal in Python

```
decimal_Number = 1028  
binValue = bin(decimal_Number)  
octalValue = oct(decimal_Number)  
hexValue = hex(decimal_Number)
```

```
print("Printing the values after the conversion of bases that is decimal to binary,octal,hexadecimal:")
```

```
print("The binary value of the given decimal number", decimal_Number, "=", binValue)
```

```
print("The octal value of the given decimal number", decimal_Number, "=", octalValue)
print("The hexadecimal value of the given decimal number", decimal_Number, "=", hexValue)
```

output:

```
The binary value of the given decimal number 1028 = 0b10000000100
The octal value of the given decimal number 1028 = 0o2004
The hexadecimal value of the given decimal number 1028 = 0b10000000100
```

Decimal to Octal using While loop in Python

```
gvn_numb = 15
a = 1
print("The Octal value of the given decimal number {", gvn_numb, "} is: ")

octl_num = 0
while(gvn_numb != 0):
    b = (gvn_numb % 8)*a
    octl_num = octl_num+b
    gvn_numb = gvn_numb//8
    a = int(a*10)

print(octl_num) -->17
```

Decimal Number to Binary in Python

```
gvn_numb = 7
a = 1

binry_num = 0
print("The Binary value of the given decimal number {", gvn_numb, "} is: ")

while (gvn_numb != 0):
    remdr = gvn_numb % 2
    gvn_numb = gvn_numb // 2
    b = (remdr * a)
    binry_num = int(binry_num + b)
    a = int(a * 10)
```

```
print(binry_num) -->111
```

Convert Binary to Decimal

```
gvn_binry_num = 1000
a = 0

deciml_num = 0
print("The Decimal value of the given binary number {", gvn_binry_num, "} is: ")

while (gvn_binry_num != 0):
    rem = gvn_binry_num % 10
    gvn_binry_num = gvn_binry_num // 10
    b = rem*pow(2, a)
    deciml_num = deciml_num + b
    a = a+1
print(deciml_num)-->8
```

Convert Octal to Decimal

```
gvn_octl_numb = 15
a = 0
deciml_numb = 0
print("The Decimal value of the given Octal number {", gvn_octl_numb, "} is: ")
while(gvn_octl_numb != 0):
    b = (gvn_octl_numb % 10)
    c = b*pow(8, a)
    deciml_numb = deciml_numb+c
    a += 1
    gvn_octl_numb = gvn_octl_numb // 10
print(deciml_numb) -->13
```